Programming for Robotics: Project Report

(Text in red in this template is provided as guidance. Please remove all red text in your final report.)

Project Title:

GitHub link:

Group Members:
- Name – Student ID
- Name – Student ID
- Name – Student ID

Specialisation Option Chosen:
☐ Embedded System (Wokwi)
☐ Autonomous Robot (Webots)
☐ Multi-Agent System (Webots)
☐ Human–Robot Interaction (Webots)

1. Project Overview

Briefly describe the goal of your project.

Include:
- What problem the robot solves
- Type of robot/system
- Key features
- Environment where it operates

Example points:
- Robot purpose
- Sensors used
- Actuators used
- Intelligent behaviour implemented

2. Technical Requirements Compliance

In this section you must explicitly explain how your system satisfies all **technical requirements** listed in the assignment sheet.

For each requirement, briefly describe how your system implements it.

This section helps reviewers verify that your system meets the required specifications.

3. System Architecture

Explain the system structure using the **Sense** → **Process** → **Decide** → **Act** model.

Explain briefly how information flows through the system.

4. Behaviour Model

Describe how the robot behaves, including:

- FSM / Behaviour Tree / Control Logic
- Minimum **4 behavioural states**
- Conditions that trigger state transitions

Explain **why this behaviour model was chosen**.

5. Perception / Sensor Processing

Explain how sensor data is used, for example:

- Sensor fusion
- Multi-condition logic
- Camera detection
- Colour tracking
- Distance measurement

Explain:

- What data is read
- How it affects decisions

6. Safety and Robustness

Explain safety mechanisms used, for example:

- Collision avoidance
- Boundary limits
- Emergency stop
- Fail-safe behaviour
- Timeout protection

Explain what happens if sensors fail or unexpected situations occur.

7. Code Structure

Briefly describe the structure of your program and explain:

- Main modules/functions
- Control loop
- Key libraries used

8. Key Code Snippets

Include important parts of your code.

Do **not paste the entire program**.
Only include sections that demonstrate key logic, such as: FSM Logic, Sensor Processing Example, etc.

Explain how the sensor influences robot behaviour.

9. Results / Demonstration Summary

Describe what the robot successfully does:

- Detects motion
- Navigates around obstacles
- Coordinates multiple robots
- Interacts with a user

Mention any limitations or challenges encountered.

10. Individual Contributions

Briefly describe what each member contributed.

Example:

Member	**Contribution**
Student A	Sensor integration, FSM logic
Student B	Webots navigation and obstacle avoidance
Student C	Video demonstration and testing